

Population status of *Acropora palmata* (Lamarck, 1816) in Cuban coral reefs

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ABSTRACT

Acropora palmata is the most important reef-building species in the crest biotopes in the Caribbean Sea. Their populations have been decimated in the last century by disease outbreaks, hurricanes and coral bleaching events. The species has been declared as threatened based on its wide-range decline and poor recovery. We evaluated the status of *A. palmata* populations across 41 shallow reefs (= sites) around the Cuban archipelago. Density, maximum diameter, and mortality percentage of *A. palmata* colonies were quantified as main metrics of status. Living coral cover (all species) and reef complexity per site were quantified as well. Metrics of *A. palmata* varied significantly among sites. *A. palmata* density was less than 5 colonies 10 m^{-1} except in one site and only three sites had dominance greater than 50% with respect to the rest of corals. The diameter of *A. palmata* colonies ranged from 10 to 600 cm, with 70% of sites having a median of less than 200 cm. The mortality was large, with 50% of the sites having more than 50% of mortality, and six sites having 100% mortality. The three sites with better apparent conditions after metric scores were determined were Faro de Cazones, Cayo Frago, and Baracoa. *A. palmata* populations had a bad or regular status in 38 out of 41 studied sites; but, the abundance of erect colonies in many of the sites favored the reef complexity and likely maintained its ecological functions. The low percentage of living coral tissue likely limits the reproductive potential of the species and hence the resiliency of reef crests. In conclusion, the overall status of the species in Cuban archipelago is bad.

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1. Introduction

Until the late 1970s, the open surfaces of Caribbean reefs were dominated by corals, algal turf, and coralline algae. These reefs displayed a generalized zonation pattern of common scleractinian coral species being primarily Elkhorn coral *Acropora palmata* (Lamarck, 1816) the primary builder of the shallow reef framework (Aronson and Precht, 2001; Roff and Mumby, 2012; Muko et al., 2013). Caribbean coral communities were historically dominated by Elkhorn corals at depths of 0–5 m and this habitat was known as the *palmata* zone (Geister, 1977; Rutzler and Macintyre, 1982). Elkhorn coral was also considered the

dominant species in the Cuban shallow reefs (Martínez-Estalella, 1986). Well-developed *A. palmata* populations have contributed to important ecological processes and services of Caribbean coral reefs such as habitat provisioning for a variety of reef organisms, coastal protection, gross community calcification, and nitrogen fixation (Alvarez-Filip et al., 2009; Roff and Mumby, 2012; Chamberl et al., 2015).

During the 1970s, a Caribbean-wide outbreak of white-band disease reduced the abundance of acroporid corals by more than 95% (Acr, 2005; Jackson et al., 2014; Mumby et al., 2014). In the following decades, the remaining Caribbean populations were subsequently impacted by hurricanes, coral bleaching, algal overgrowth, and predation by corallivorous snails and damselfishes (Williams and Miller, 2012; Rogers and Miller, 2013; Ricaurte et al., 2016). The relatively rapid loss of *A. palmata* has been

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unprecedented in the fossil record (Perry et al., 2015). Therefore, the species is classified as threatened under the U.S. Endangered Species Act of 1973, and is included in the Red List of Threatened Species of 2008 as *critically endangered* (IUCN, 2011) as well. The decline of *A. palmata* significantly reduced the living coral cover and the resilience of the reef crests because it is the most important reef-building species in the Caribbean (Pandolfi and Jackson, 2006).

The recovery of *A. palmata* populations in the Caribbean region have shown positive signs in some locations based on the occurrence of recruitment, persistence of healthy and potentially reproductive colonies, growing, and resheeting on relicts of *Acropora* skeletons (Bruckner, 2002; Bonito, 2006; Zubillaga et al., 2008; Larson et al., 2014). Population assessments of *A. palmata* have been performed in the U.S. Virgin Islands and in the Florida Keys, reporting in overall a low living coral cover (e.g. Miller et al., 2002; Mayor et al., 2006). Other studies in the Mesoamerican Reef System (Rodríguez-Martínez et al., 2014) and in Venezuela reefs (Croquer et al., 2016) have reported little or no recovery as well. In Cuba, two studies have reported the spatial trends of *A. palmata* in crests at large spatial scales in the archipelagos of Sabana-Camaguey (González-Ferrer et al., 2007) and Canarreos (Alcolado et al., 2010). The results of these studies using the AGRRA methodology indicated that the general condition of reef crests was poor; however, *A. palmata* was not the targeted species. Recently, a study by Hernández-Fernández et al. (2019) was published about the status of *A. palmata* in the reef system Jardines de la Reina, which was evaluated as one of the best-preserved systems in the Caribbean. The general condition of the populations was evaluated between good and not so good in the study area (Hernández-Fernández et al., 2019).

The goal of our study was to assess the status of *A. palmata* populations around the Cuban archipelago, highlighting those reefs in better or worse condition according three metrics analyzed in an integrative approach. The results constitute an updated baseline of the species population status and provide a basis for hypotheses about regional tendencies of decline and recuperation of the populations. We have surveyed a large fraction of the *A. palmata* crests in the Cuban Archipelago (i.e. at least more than 50% of the reefs where populations occur) giving a large relevance to our study for the regional knowledge about the species.

2. Materials and methods

2.1. Studied reefs

The study included 41 sites across the Cuban Archipelago (Fig. 1). The sites were chosen and classified as representative of *A. palmata* crests according to local knowledge and previous studies (Caballero-Aragón and Perera-Valderrama, 2014; Caballero-Aragón et al., 2019). All of the selected crest reefs had a continuous reef tract with an extension of at least 300 m along corresponding isobaths and depth range of 1–3 m. The reefs were located far away from contamination sources and large urban settlements (except BAR and SFE, located near to Havana City). Overall, most of the sites could be roughly considered as “relative clean reefs respect to terrestrial contamination sources” (Caballero-Aragón et al., 2019). The distance between two adjacent sites was at least 1 km. The names and geographical coordinates of the sites are given in Table 1. The sites were sampled between 2010 and 2012 during the summer months.

2.2. Sampling methodology

The sampling unit was a 10 m long linear transect (Kramer and Lang, 2003). At each site, we made visual observations using SCUBA diving gear. We haphazardly set between 10 and 22 line transects on the reef, always on the windward side of the crest and starting in one end. The number of transects per site varied according to reef occurrence, sampling restrictions and extension of the system.

We quantified in each transect three metrics of the reef (but not specifically on *A. palmata*):

- Coral colony density (colonies 10 m^{-1}): Number of colonies of all coral species (larger than 10 cm diameter) within a 10 m transect; the counts were made just under the transect tape.
- Living coral cover (%): Percent of living coral (all the species) that intercepted each transect.
- Reef complexity (cm): The averaged vertical distance between highest and lowest points in the bottom relief (included the corals) in an area of one meter of diameter. Five measures were made along each lineal transect each 2 m (i.e. 0, 2, 4, 6, and 8 m from the starting point).

We quantified in each transect four metrics for *A. palmata* population:

1. Relative abundance of *A. palmata* (%): It was calculated by dividing *A. palmata* colony density (colonies larger than 10 cm diameter under the tape) by total colony density (i.e. all the coral species registered in the reef) and multiplying it by 100.
2. Coral colony diameter (cm): Maximum lineal diameter of *A. palmata* colonies, measured from top view and including dead parts. The frequency of *A. palmata* colony diameters was represented using three size classes: small (10–50 cm), medium (51–200 cm), and large (greater than 200 cm) colonies.
3. Mortality (%): Percent of dead coral tissue in each colony of *A. palmata*. The frequency of mortality per colony was represented using three classes: low (less than 25%), medium (25%–75%), and high mortality (76%–100%).

Standing dead colonies of *A. palmata* (identifiable to the species level due to its typical morphology) were included in the analysis of density and size. Totally dead colonies which were very eroded or covered by other organisms were not included.

2.3. Data analysis

The density of *A. palmata* per site was presented in a scatter plot graph using the median as the measure of central tendency (Weissgerber et al., 2015). Also, coral colony diameters and mortality of *A. palmata* were represented with box-plot graphics using the median as measure of central tendency as well. Relative values of abundance and frequencies of coral colony diameters and mortality were represented at each site using bar graphs. For the analyses and graphical representation of *A. palmata* colony diameter and mortality, we included just the sites with more than 10 colonies; the justification is to give estimates of population metrics for representative sites of crest and not merely for sites that used to be *A. palmata* crests but are not anymore (see discussion below).

To determine the existence of significant differences between sites, with respect to density, diameter, and mortality of *A. palmata*, statistical significance was tested with PERMANOVA test (Anderson et al., 2008) using “site” as a fixed factor (i.e. one-way factorial design) and 9999 permutations. Calculations were made in the software PRIMER 6.1.15 (Clarke and Gorley, 2006) and PERMANOVA+ (Anderson et al., 2008). Bivariate correlations between the five condition metrics (abundance, density and diameter of *A. palmata*, plus living coral cover and reef complexity)

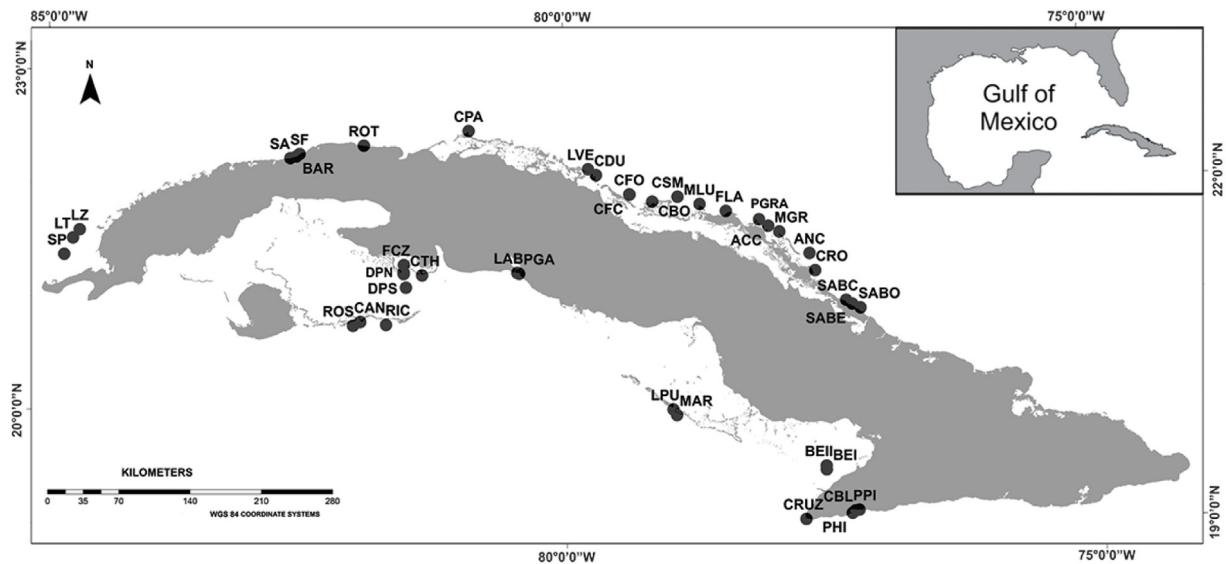


Fig. 1. Map of the sites in the Cuban Archipelago. The names and abbreviations of sites are given in Table 1.

Table 1

The study sites with names, abbreviations, geographical location, and sampled years.

Site	Name	Abbreviation	Latitude N	Longitude W	Sampled year
1	Sancho Pardo	SP	22.1610	−84.7495	2010
2	La Tabla	LT	22.3071	−84.6695	2010
3	La Zorrita	LZ	22.3809	−84.6075	2010
4	El Salado	SA	23.0409	−82.6074	2010
5	Baracoa	BAR	23.0567	−82.5503	2010
6	Santa Fé	SF	23.0788	−82.5183	2010
7	Rotilla	ROT	23.1567	−81.9041	2010
8	Cruz del Padre	CPA	23.2885	−80.8981	2012
9	Cayo La Vela	LVE	22.9467	−79.7558	2012
10	Cayo Dutton	CDU	22.8939	−79.6794	2012
11	Cayo Borracho	CBO	22.6533	−79.1453	2012
12	Cayo Fragoso oeste	CFO	22.7203	−79.3631	2012
13	Cayo Fragoso centro	CFC	22.7183	−79.3603	2012
14	Cayo Santa María	CSM	22.6936	−78.9025	2012
15	Cayo Ancilitas	ACC	22.4250	−78.0408	2012
16	Cayo Flamenco	FLA	22.5625	−78.4431	2012
17	Cayo Media Luna	MLU	22.6267	−78.6931	2012
18	Paredón Grande	PGRA	22.4825	−78.1272	2012
19	Cayo Mégano Grande	MGR	22.3719	−77.9369	2012
20	Cayo Antón Chico	ANC	22.1769	−77.6528	2012
21	Cayo Romano	CRO	22.0247	−77.6025	2012
22	Cayo Sabinal oeste	SABO	21.6878	−77.1817	2012
23	Cayo Sabinal centro	SABC	21.7578	−77.3158	2012
24	Cayo Sabinal este	SABE	21.7228	−77.2589	2012
25	Cayo Rico	RIC	21.5753	−81.6837	2010
26	Cayo Cantiles	CAN	21.5992	−81.9275	2010
27	Cayo Rosario	ROS	21.5650	−81.9968	2010
28	Diego Pérez norte	DPN	22.0379	−81.5173	2012
29	Diego Pérez sur	DPS	22.0237	−81.5173	2012
30	Faro de Cazones	FCZ	22.1043	−81.5159	2012
31	Cayo Thaelman	CTH	22.0125	−81.3417	2012
32	Laberinto	LAB	22.0344	−80.4370	2012
33	Punta Gavilán	PGA	22.0266	−80.4187	2012
34	La Puntica	LPU	20.8167	−78.9667	2012
35	Mariflores	MAR	20.7667	−78.9333	2012
36	Buena Esperanza I	BEI	20.2667	−77.5333	2012
37	Buena Esperanza II	BEII	20.3000	−77.5333	2012
38	Cabo Cruz	CRUZ	19.8333	−77.7333	2012
39	Punta Hicacos	PHI	19.8794	−77.2993	2012
40	Punta Piedra	PPI	19.9047	−77.2354	2012
41	Cayo Blanco	CBL	19.8972	−77.2746	2012

were analyzed with non-parametric Spearman rank correlation, using medians per site as input values and $p = 0.05$ as threshold of statistical significance.

The qualitative status of *A. palmata* populations was assessed on basis of the three metrics explained above: (1) relative abundance, (2) frequency of large coral colonies, and (3) complement of mortality (i.e. $100 - \text{mortality}$). Three categories of status

were arbitrarily defined: bad (0–24), regular (25–75), and good (75–100).

3. Results

The median of *A. palmata* colony density for Cuban archipelago was 1 colony 10 m^{-1} (range: 0–13 colonies 10 m^{-1}). The median density per site never exceeded 5 colonies 10 m^{-1} except in FCZ and there were significant differences among sites ($p = 0.0001$). The distribution of coral density was largely asymmetric with 70% of sites presenting median values less than or equal to 1 colony 10 m^{-1} (Fig. 2A). The median of *A. palmata* relative abundance across the 41 sites was 22% (range: 0–81%). Only 3 sites had values greater than 50%, and FCZ presented the highest dominance of *A. palmata* (81%). The rest of the sites had values well below 40%, and in four sites the species was absent (Fig. 2B). Density and abundance of *A. palmata* were positively and significantly correlated (Table 2).

Only 27 sites, out of 41, had more than 10 colonies of *A. palmata*. Consequently, coral colony diameter, mortality, and living coral tissue analyses were made using this sampling size. Coral colony diameter had a median across the 27 sites of 120 cm (range: 10–590 cm) and varied significantly among sites ($p = 0.0001$). Seventy percent of sites presented median coral colony diameter smaller than 200 cm and just one (CTH) had median coral colony diameter larger than 240 cm (Fig. 3A). The analysis by coral colony diameter classes indicated greater frequency of intermediate diameters (i.e. 51–200 cm). CTH had high frequency of large colony diameter (larger than 200 cm) and no young colonies (i.e. 10–50 cm). The sites with higher frequency of small colonies were ROT, CRO, and CPA (Fig. 3B). The coral colony diameter distribution was shown in ESM Appendix 1A.

The median of *A. palmata* mortality across the 27 sites was 45% (range: 0–100%) and varied significantly ($p = 0.0001$). CPA, FCZ, and CRO presented median mortality less than 10%. Half of the sites had a median mortality higher than 50%, and six sites had 100% (Fig. 4A). The analysis by mortality classes indicated that eight sites showed a frequency larger than 50% of healthy colonies (i.e. less than 25%) and six sites had 50% of their colonies with 100% of mortality (Fig. 4B). The mortality distribution was shown in ESM Appendix 1B. Mortality was significantly correlated (positively) with coral colony diameter and reef complexity. The two latter variables were also significantly correlated (Table 2). The values of living coral cover and reef complexity was shown in ESM Appendix 2.

Only one site could be qualified as good according to relative abundance and frequency of large colonies; and only 10 sites were classified as good for mortality (Table 3). Most of the sites were included in the categories of bad or regular, indicating that the overall status of *A. palmata* Cuban populations is bad.

4. Discussion

Here we report data about the trends of fundamental reef metrics (e.g. density, mortality, complexity). We also provide an integrated analysis to reach conclusions about the overall status of *A. palmata* populations at scale of the whole Cuban archipelago. Our discussion is structured in three parts: First, we discuss the values of the reef metrics in a regional context (Florida and Caribbean). Second, we present the general pattern emerging of the integration of results with an assessment of the number of sites in bad, regular or good conditions. Third, we consider the possible causes of these patterns in the context of global coral reef decline.

The *A. palmata* density in coral reefs from Florida Keys and the Caribbean Sea has been low since the beginning of the twenty-first one century. In the Florida Keys, the species was aggregated in specific habitats and density was well below 1 colony m^{-2} (Bruckner, 2002; Acr, 2005). In Banco de Chinchorro Reef (Mexico), densities ranged from 0.11 to 0.44 colonies m^{-2} (Vega-Zepeda et al., 2007), while in the Veracruz Reef System, ranged from 0.02 to 0.28 colonies m^{-2} (Larson et al., 2014). Low coral colony density values for *A. palmata* have been reported also for St. Croix (0.02 colonies m^{-2}), St. John (0.06 colonies m^{-2}), and Venezuela (0.08 colonies m^{-2}) (Mayor et al., 2006; Schärer et al., 2008; Zubillaga et al., 2008). The *A. palmata* density in Cuban reefs seems comparatively higher than in some neighboring countries although our results have been expressed in different units due to the sampling method (i.e. linear transect).

Coral colony diameter and the frequency of large diameter classes of *A. palmata* were comparatively higher in our study with respect to regional assessments. For example, crests from Veracruz reported coral colony diameter of $58 \pm 73\text{ cm}$ (Larson et al., 2014). In Florida, young colonies (0–50 cm) currently prevailed because large colonies with diameter range of 100–500 cm were mostly dead or very eroded (Vardi et al., 2012). Equally, in our study, colonies of *A. palmata* remained standing up despite widespread mortality of coral tissue. Indeed, the reef flattening that has occurred in Caribbean reefs (Alvarez-Filip et al., 2009, 2013) was not widespread in Cuba even for those sites with 80–100% mortality of *A. palmata*. The occurrence of many large and erected colonies suggests that *Acropora* crests still provide ecological functions to coastal ecosystems (e.g. structural complexity and shelter, dissipation of wave energy). Furthermore, the permanence of many erected colonies could favor resheeting events, which have been previously reported in Canarreos reefs of Cuba (Alcolado et al., 2010) and in the Florida Keys (Bonito, 2006).

The relatively high frequency of large colonies in many sites in the Cuban archipelago suggests a moderate to high spawning potential because of the positive relationships between the coral size and the reproductive potential (Jordan-Dahlgren, 1992). However, despite large median colony diameter, the proportion of dead coral tissue was rather high (except for FCZ). The other two sites with potentially high spawning potential were CFO and BAR more likely because of the greater colony density than by the colony diameter. The reproduction of *A. palmata* is by asexual fragmentation or by liberation of gametes to the environment (Szmant, 1986). Likely, the reproductive potential of *A. palmata* in most of the sites was limited due to the low percentage of living coral tissue, resulting in a large limitation for the recovery of populations. Pilot studies of recruitment realized in some of the studied sites (DPN, DPS, LAB, and PGA) did not report more than 1 recruit m^{-2} of *A. palmata* (Caballero-Aragón and Perera-Valderrama, 2014). This situation favors to other species with brooding strategies, such as *Porites astreoides* (Lamarck, 1816) or those with short larval periods such as *Millepora complanata* (Lamarck, 1816) (Aronson and Precht, 2001).

According to the integrated analysis, we recorded good condition of *A. palmata* population in FCZ, CFO, and BAR. They were the only sites with a relative abundance of *A. palmata* over 50%. Sites with bad or regular condition of *A. palmata* populations were PGA, LPU, CTH, MAR, and CAN. Despite these sites have intermediate values of coral density and large sized colonies, the high mortality of coral tissue suggests these reefs have experienced significant environmental stress. The 14 crests that were not included in the analyses of colony diameter and mortality were in worse condition mainly because very low density of colonies (i.e. less than 10 colonies per reef). These sites were dominated by isolated dead colonies giving a low chance of recovery. Actually, these

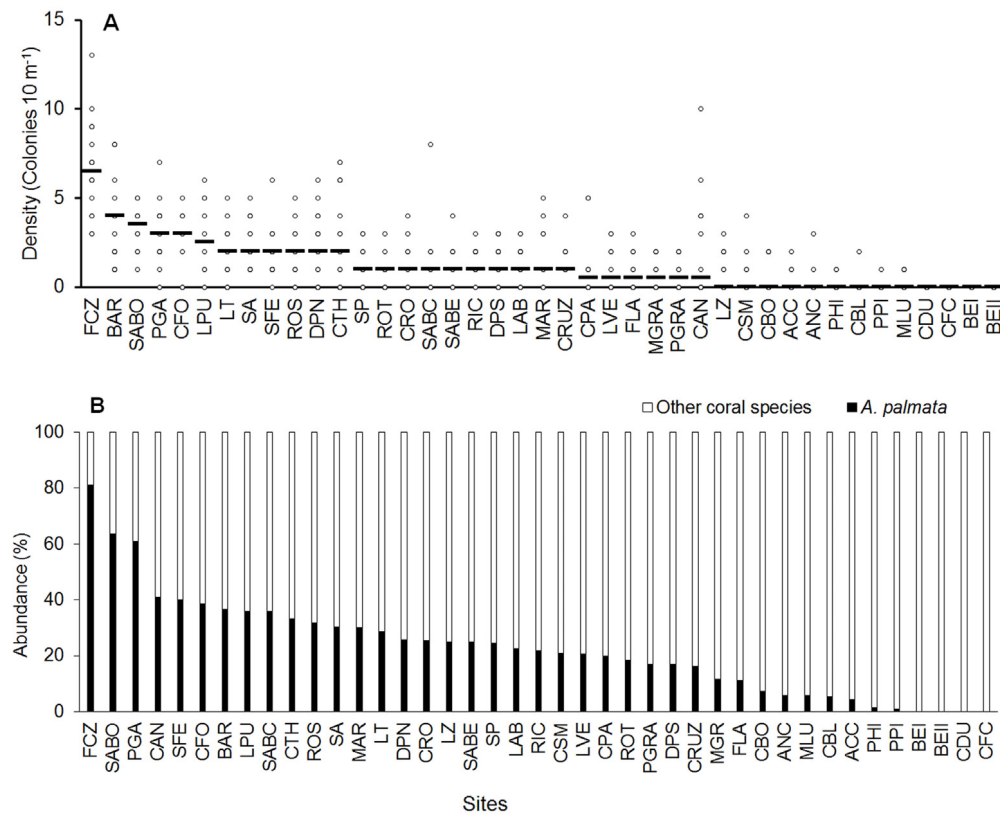


Fig. 2. Abundance of *A. palmata* across the sites. (A) The *A. palmata* colonies density represented as medians (bars) and observations (open circles). (B) Relative abundance of *A. palmata* respect to the total abundance of corals. Data ordered by abundance values. Full names of the sites are given in Table 1.

Table 2

Bivariate correlations between six variables analyzed with non-parametric Spearman rank correlation, using the median per site ($N = 27$) as input values and $p = 0.05$ as cut-off value for significant correlations (in bold). Apal = *A. palmata*.

	Living coral cover	Apal abundance	Apal colony density	Apal colony diameter	Apal mortality
Reef complexity	0.28	0.29	0.32	0.47	0.43
Living coral cover		−0.03	0.26	0.30	−0.19
Apal abundance			0.72	0.33	0.29
Apal colony density				0.26	0.25
Apal colony diameter					0.57

Table 3

Assessment of the status of *A. palmata* populations based on three metrics measured across sites and three qualitative categories. The number of sites within each category is given in the cells. Note that 41 sites were assessed for relative abundance (metric 1); but only 27 sites had colonies larger than 10 cm diameter and therefore used for metrics 2 and 3. Note also that mortality was expressed as the complement to be assessed in the same scale as the metrics 1 and 2.

Metric	Bad (0%–24%)	Regular (25%–75%)	Good (76%–100%)
1. Relative abundance	22	18	1
2. Frequency of large colonies	17	9	1
3. (100 – Mortality)	9	8	10

sites should not be considered as *A. palmata* crests anymore, but rather as rocky shallow reefs with certain growth of small incrusting corals and gorgonians.

A. palmata was poorly correlated with reef complexity and living cover indicating a decline in the relative contribution of the species to two important features of the coral reef community: rugosity and living coral cover. Indeed, two species dominated those crests with bad or regular condition: *M. complanata* and *P. astreoides*. *M. complanata* has a short larval period and can proliferate in coral reefs impacted by overfishing, diseases, and physical

damage (Lewis, 1989). *P. astreoides* has a rapid generation, availability of self-fertilization, and capacity to release mature and larger larvae (Doropoulos et al., 2015). The adult corals of *P. astreoides* are tolerant to bleaching, hurricanes, and sedimentation (Lirman and Fong, 1996, 1997; Torres and Morelock, 2002). A phase shift, from *A. palmata*-dominated reef to *M. complanata* + *P. astreoides*-dominated reef, has likely occurred in many reef crests of Cuban archipelago. This phase shift has probably reduced the reef complexity and limited the space available for *A. palmata* recruitment (Alcolado et al., 2010).

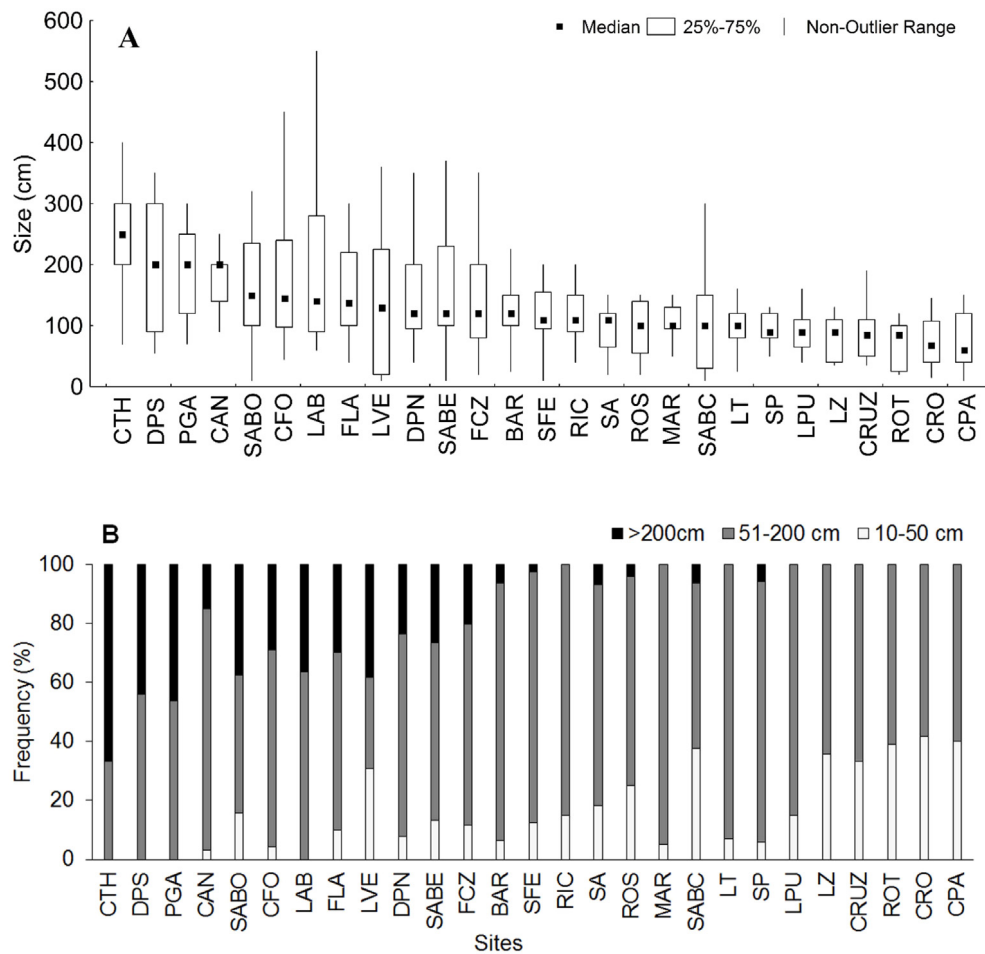


Fig. 3. Coral size of *A. palmata* across the sites. (A) Coral colony diameter (size). (B) Relative distribution by diameter classes. Data ordered by coral size values. Full names of the sites are given in Table 1.

The *A. palmata* populations had large spatial variability reflecting a high level of environmental stress (Warwick and Clarke, 1993). The influence of adverse natural and anthropogenic drivers acting at local and regional spatial scales likely explains the increased variability, compromised resilience, and the general trend of reef decline in Cuban archipelago (González-Díaz et al., 2018; Caballero-Aragón et al., 2019). However, the accurate identity and relative contribution of these ecological drivers is challenging because the poor knowledge of the environmental evolution in most of the Cuban reefs and the lack of key data (e.g. nutrients, chlorophyll, pathogens). This variability likely responded to the modern influence of stressors such as warming, disease outbreaks, and hurricanes (Harborne et al., 2017). Pollution and eutrophication derived from agriculture and coastal development have had a rather limited negative influence on the studied coral reefs in Cuba. This may reflect the resilience of affected reefs or the scarcity of relevant measurements (e.g. nutrients, sedimentation). For instance, in two sites close to Havana City (SA and SF) no organic contamination was reported on basis of low concentrations of coprostanol (0.03 and $0.02 \mu\text{g g}^{-1}$, respectively) in surface sediments (Castellanos-Iglesias et al., 2018; Martins et al., 2018). The other site near to Havana City (BAR) reported mild effect of eutrophication based on ^{15}N isotopic analysis of the macroalgae *Sargassum hystrix* (Durán et al., 2018). Nevertheless, the development of the private property, tourism, mining, and subsistence fishery will yield additional local impacts in Cuban reefs. Despite the importance of these local impacts, there are not consistent measurements of them mainly due to logistic

limitations. Reinforcement of the environmental protection in Cuba (e.g. creation of marine protected areas) in the last three decades has had a limited impact on the recovery of *A. palmata* maybe pointing to the principal role of global stressors on the reefs (Bruno et al., 2019).

5. Conclusions

The structure of Cuban *A. palmata* populations was characterized by large spatial variability, likely reflecting a high level of environmental stress, similar to that reported in the Great Caribbean region. The status of *A. palmata* population was good just in three sites, and bad or regular in 38 sites because the low density of colonies and the high percentage of mortality. Actually, at least 14 sites should not be considered as *A. palmata* crests anymore, but rather as rocky shallow reefs with certain growth of small incrusting corals and gorgonians. Despite this overall negative status, the abundance of colonies still erect maintains the reef complexity in many sites. The low percentage of living coral tissue in most of the sites limits the reproductive potential of the species and hence the resiliency of reef crests. The deterioration of *A. palmata* populations has likely been due to the combination of global stressors (e.g. warming, diseases, and hurricanes) that has exacerbated fundamentally during the last three decades, and anthropogenic local stressors (e.g. overfishing, pollution).

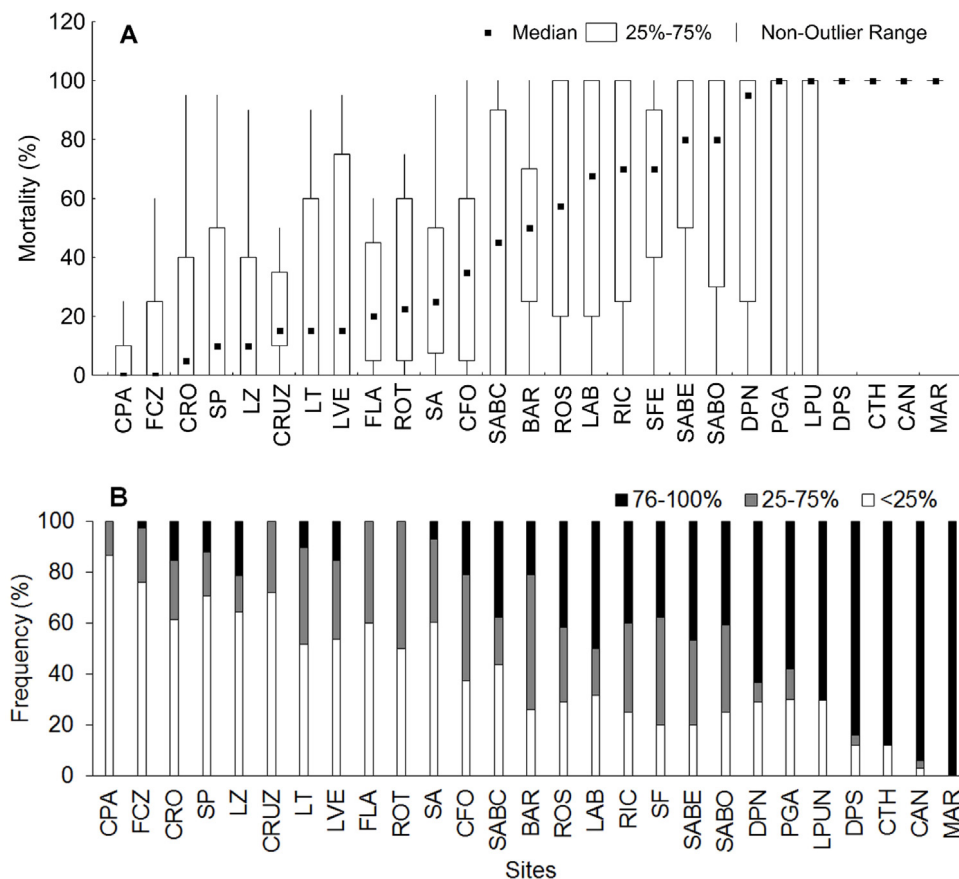


Fig. 4. Mortality of *A. palmata* across the sites. (A) Mortality. (B) Relative distribution by mortality classes. Data ordered by mortality values. Full names of the sites are given in Table 1.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Hansel Caballero-Aragón: Funding acquisition, Writing - original draft. **Susana Perera-Valderrama:** Funding acquisition, Writing - review & editing. **Néstor Rey-Villiers:** Writing - review & editing. **Juliett González-Méndez:** Funding acquisition, Writing - review & editing. **Maickel Armenteros:** Supervision, Writing - review & editing.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.rsma.2019.101029>.

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